

FEYZULLAH YAVAN

✉ feyzullah.yavan@kit.edu 📞 +49 152 26167530



Date of Birth: 20.04.1998
Linked-in Profile: www.linkedin.com/in/ugurfey
GitHub Profile: <https://github.com/ugurfeyzullah>
Marital status: Single

EDUCATION

2024 – Present	Department of Civil Engineering, Geo and Environmental Sciences (Doctoral Candidate) KIT- Karlsruhe Institute of Technology	Karlsruhe, Germany Average grade: -
2022 – 2022	Erasmus Exchange Program KIT- Karlsruhe Institute of Technology	Karlsruhe, Germany Average grade: 4.00/4.00
2021 – 2023	Department of Civil Engineering (Master's Degree) KTÜ- Karadeniz Technical University	Trabzon, Turkey Average grade: 3.93/4.00
2018 – 2019	Erasmus Exchange Program VSB-Technical University Of Ostrava	Ostrava, Czechia Average grade: 3.40/4.00
2016 – 2021	Department of Civil Engineering (Bachelor's Degree) Sakarya University	Sakarya, Turkey Average grade: 3.61/4.00

RESEARCH EXPERIENCE AND PUBLICATIONS

1. AI-Based Design Optimization in BIM Projects

- *Structural Optimization of Trusses in Building Information Modeling (BIM) Projects Using Visual Programming, Evolutionary Algorithms, and Life Cycle Assessment (LCA) Tools* – <https://doi.org/10.3390/buildings14061532>
- *Reliability-Constrained Structural Design Optimization Using Visual Programming in Building Information Modeling (BIM) Projects* – <https://doi.org/10.3390/app15031025>
- *Enhancing Spatial Truss Designs by Integrating Metaheuristic Optimization Techniques via Visual Programming in BIM-based Projects* – <https://doi.org/10.3390/app1503102>

2. Field Information Modeling

- *Sketch-to-BIM: A comparative analysis for automated truss design in Dynamo* – https://doi.org/10.31462/icearc2025_ce_mng_736
- *Sketch-To-Bim: Enhancing The Dynamo Based Automated Truss Design Optimization Workflow* – <https://doi.org/10.22260/CCC2025/0076>
- *Sketch-to-BIM: A Unified AI Framework for Design Automation and Optimization from Sketch, Speech, and Text in Dynamo* – <https://dx.doi.org/10.2139/ssrn.7116288>
- *Automatic alignment of point clouds onto building information model (BIM) in regular rectangular buildings* – <http://dx.doi.org/10.31462/icearc.2023.cme960>
- *Scan-to-BIM: Semantic 3D Reconstruction from Registered Point Clouds and 360-Degree Panoramas for AEC sector (Ongoing)*
- *Integrating Augmented Reality with Ultra-Wide Band Positioning for Improved Accuracy in Construction Applications (Ongoing)*
- *Modeling a Real-World Project in Revit with LOD 300 and Performing Energy Analysis Using the Green Building Studio Plug-In: A Comparative Study of Results with TS825 Regulation Standards (Bachelor thesis)*

3. Generative Design

- *Simultaneous Design and Management Optimization through AI-Based Metaheuristic Algorithms and Generative Adversarial Networks. (Ongoing)*
- *Natural-Language/Voice-Driven Architectural Modeling in Revit with Simultaneous Plan Synthesis, Material Assignment, and Fabrication-Safe Model Extraction (Ongoing)*
- *Design, 3D Printing and Analysis of Vertical Hydroponic Tower for Sustainable Development (Erasmus project)*

4. Digital Lean Management Tool Generation

- *A Flask-based Website for Project Planning and Notes Management System with Kanban Board, Team Key Validation, and Chatbot Integration powered by LLM.* – <https://kitdec.pythonanywhere.com/>
- *A Flask-based Website for Project Management Assistant and AI-Powered Guidance.* – <https://kitdec2.pythonanywhere.com/>

PROFESSIONAL EXPERIENCE

02/2024 – Present	Research Assistant at Digital Engineering and Construction		
	<i>Karlsruhe Institute of Technology</i>	Supervisor: <i>Reza Maalek</i> <i>Shervin Haghsheno</i>	<i>Karlsruhe/Germany</i>
02/2022 – 1/2024	Research Student at Digital Engineering and Construction		
	<i>Karlsruhe Institute of Technology</i>	Supervisor: <i>Reza Maalek</i>	<i>Karlsruhe/Germany</i>
01/2021 – 06/2021	Civil Engineer at General Directorate of Construction		
	<i>Sakarya University of Applied Sciences</i>	General Manager: <i>Osman Şahin</i>	<i>Sakarya/Turkey</i>

SKILLS

Language	Tools & Technologies	
• Turkish (Mother tongue) • English (C1) • German (B1)	• Fusion 360 • Navisworks • SAP2000 • Dynamo • Adobe illustrator • IdeCAD	• Revit • Robot Structural Analysis • MS Office • AutoCAD • Python • MATLAB

SYMPOSIUMS & CONFERENCES

2025	35 th BBB Tagungs (Participant)	Stuttgart, Germany
2025	CCC 25 (Conference Speaker)	Zadar, Croatia
2025	ICEARC 25 (Conference Speaker)	Trabzon, Turkey
2024	IASS 24 Computational Methods Symposium (Conference Speaker)	Zurich, Switzerland
2023	DIGISUSTAIN 23 (Participant)	Frankfurt, Germany
2023	ICEARC 23 (Conference Speaker)	Trabzon, Turkey

REFERENCES

Reza Maalek	<i>KIT - Karlsruhe Institute of Technology</i>	Professor	+49 721 608 44350	reza.maalek@kit.edu
Dominik Waleczko	<i>KIT - Karlsruhe Institute of Technology</i>	Research Assistant	+49 721 608-48265	dominik.waleczko@kit.edu
<i>Sedat Semih Çağlayan</i>	<i>SUBU - Sakarya University of Applied Sciences</i>	Assistant Professor	+90 0505 866 96 62	semihcaglayan@subu.edu.tr
<i>Vedat Toğan</i>	<i>KTU - Karadeniz Technical University</i>	Professor	+90 0533 430 98 35	togan@ktu.edu.tr

*Upon request, recommendation letters or an Arbeitszeugnis will be provided.

EXTRACURRICULAR ACTIVITIES

2022 – Present	Open-Source Apps Development	•	Creating apps for Learning & Productivity
2021 – 2023	KTÜ Paragliding Community	•	Membership and volunteer works
2019 – 2021	SUBÜ Traditional Sports Community	•	Founder and Manager
2017 – 2021	SAU Horse Riding Club	•	Membership and volunteer works
2016 – 2021	SAU Traditional Turkish Archery Community	•	Work of an educator